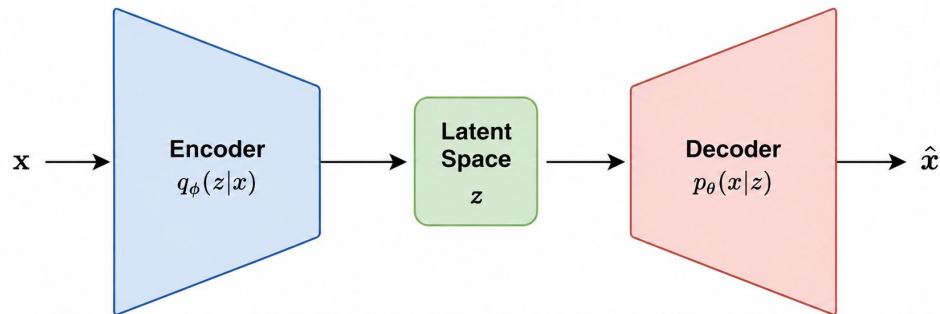


VAE's are a natural choice to generate a continuous latent space



Continuity of the latent space, allows an RG scheme, akin to real space RG formulation.

By identifying hierarchical embedding of nodes across scales, we identify the flow of probabilities of edges.

$$p_{\theta}(e_k = (u, v) \mid \mathcal{E}_{k-1}, Z) = \frac{\exp(\theta(z_u, z_v))}{\sum_{(u', v') \in \mathcal{E}_{k-1}} \exp(\theta(z_{u'}, z_{v'}))}$$

$$P_{\theta}(G \mid Z) = \prod_{k=1}^{|E|} p_{\theta}(e_k \mid \mathcal{E}_{k-1}, Z)$$

Loss:

$$\max_{\theta} \max_{\phi} \mathbb{E}_{\mathcal{D}} \left[-\text{KL}(q_{\phi}(z|x) \parallel p(z)) + \mathbb{E}_{q_{\phi}(z|x)} \log p_{\theta}(x|z) \right].$$

Renormalisation scheme

Proximity measure : The nodes are gaussian embedded in the latent space. The proximity of two nodes is the **Battacharya distance** of their gaussians.

$$D_B = \frac{1}{8}(\mu_1 - \mu_2)^T \Sigma^{-1}(\mu_1 - \mu_2) + \frac{1}{2} \ln \left(\frac{\det(\Sigma)}{\sqrt{\det(\Sigma_1) \det(\Sigma_2)}} \right)$$
$$\Sigma = \frac{\Sigma_1 + \Sigma_2}{2}$$

Node grouping : connected clusters, in which the proximity measure between each pair of nodes is **within a certain threshold**, is grouped to form a supernode.

Coarse Grained graph : The edges of the supernode is **identified trivially** based on the edges of its children.

Renormalisation :

The flow of probabilities of edges is identified such that the **lnP(G)** is equal to the **reconstruction error** associated with the embedding of the empirical graph.

Renormalisation scheme

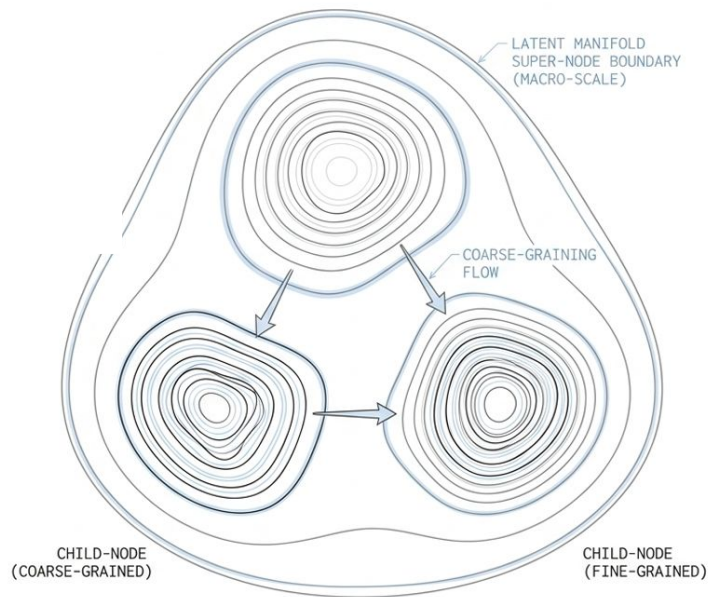
Embedding the super node : the supernode is embedded such that the collective proximity measure between the supernode and it's children nodes is maximized.

$$z_S^* = \arg \min_{z_S} \sum_{i \in \mathcal{C}(S)} D_B(z_S, z_i)$$

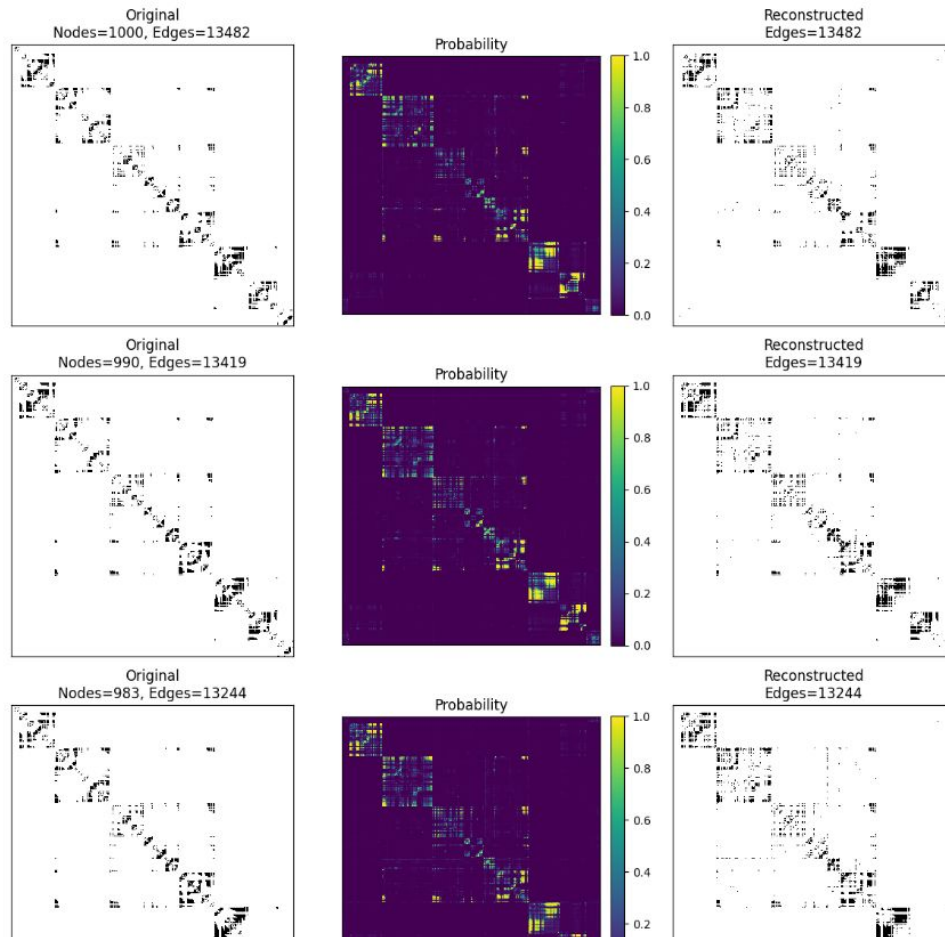
The **standard deviation** of such a supernode can be identified as the **coarse grained scale** (local, for it is specific to that region of the latent space where the supernode is embedded) in the latent space.

This way, we identify **hierarchical embedding of nodes across scales**.

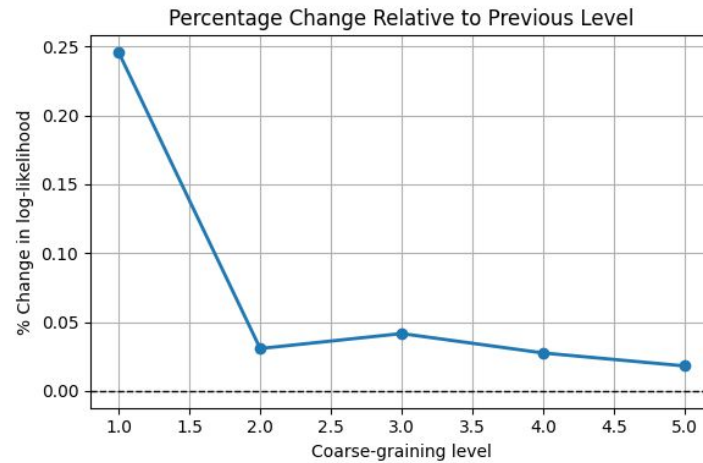
The flow of probabilities also represents the **likelihood of an interaction between different regions of the latent space across scales**.



The RG scheme is consistent across scales



The renormalisation condition across scales



Invariance of the scale-free property of a natural visibility network, Constructed from a chaotic time series

